

SEMICONDUCTOR INDUSTRY REPORT

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1 EXECUTIVE SUMMARY

Industry: Semiconductor | GICS Code: 45301020 | Sub-Industry: Semiconductors

Current Stage: High-growth structural upcycle — the fastest and most sustained demand cycle in the industry's history, driven by AI infrastructure buildout rather than cyclical consumer demand.

Market Size & Growth:

- 2025 actual: \$793B (+21% YoY)
- 2026 forecast: ~\$975B (+26% YoY)
- 2030 target: ~\$1.3–1.5T
- 2036 long-term outlook: ~\$2T
- Near-term CAGR (2026–2030): ~11–13% | Long-term CAGR (2026–2036): ~7.5%

Key Investment Thesis:

The global semiconductor industry has undergone a structural transformation — transitioning from a cyclical hardware business into the foundational infrastructure layer of the AI economy. With AI chips accounting for approximately 50% of 2026 industry revenue despite representing less than 0.2% of total unit volume, value is concentrating rapidly at the leading edge of the supply chain. This creates a generational investment opportunity in companies that control the most irreplaceable nodes of the semiconductor value chain — particularly advanced foundry manufacturing, AI chip design, and leading-edge packaging — where competitive moats are wide, switching costs are extreme, and government support provides an additional layer of downside protection.

Major Risks:

- **Taiwan concentration** — 92% of advanced node chips produced in one geopolitically exposed location; no near-term alternative at leading-edge nodes
- **AI demand concentration** — ~50% of industry revenue dependent on one end market; hyperscaler AI capex slowdown would cause sharp revenue contraction
- **China materials weaponisation** — dominant control of gallium, germanium, and tungsten with no Western alternative supply available in the near term

2 INDUSTRY OVERVIEW (45301020)

In the modern era of technology, semiconductors are the foundational building block of virtually all electronic devices and the global digital economy, allowing devices to process, store, and transmit information. AI demand has transformed the industry's revenue structure, with AI chips now contributing approximately 50% of total industry revenue. The market is forecast to grow 26% in 2026 alone, reaching approximately \$975 billion, with a longer-term CAGR of roughly 7.5% expected to carry the industry toward \$2 trillion in annual revenue by 2036.

Major Products

The industry produces logic chips (CPUs, GPUs, AI accelerators), memory chips (DRAM, NAND), Analog chips, power semiconductors, and specialty chips for automotive, industrial, and defence applications.

Segments in industry

The industry is divided into 3 major segments – Fabless, Foundries and OSAT

- **Fabless** - design only, outsource manufacturing completely. their industry share is ~35%
Eg: NVIDIA, Apple, AMD, Qualcomm
- **Foundries** - manufacture only they don't design the chips, their industry share is ~20%
Eg: TSMC, SMIC, Global foundries
- **OSAT** - responsible for outsourced packaging and testing, their industry share is ~7%
Eg: ASE group, Amkor, JCET
- **IDM** - design, manufacture and package all in-house, their industry share is ~38%
Eg: Intel, Samsung, Texas Instruments

Industry life cycle

Semiconductor industry is currently in the growth stage of its industry life cycle, driven by rapid adoption of AI, cloud computing, 5G networks, EVs, and Internet of Things (IoT) applications. Demand for advanced chips continues to outpace supply, encouraging significant investments in new fabrication facilities and R&D.

Value chain

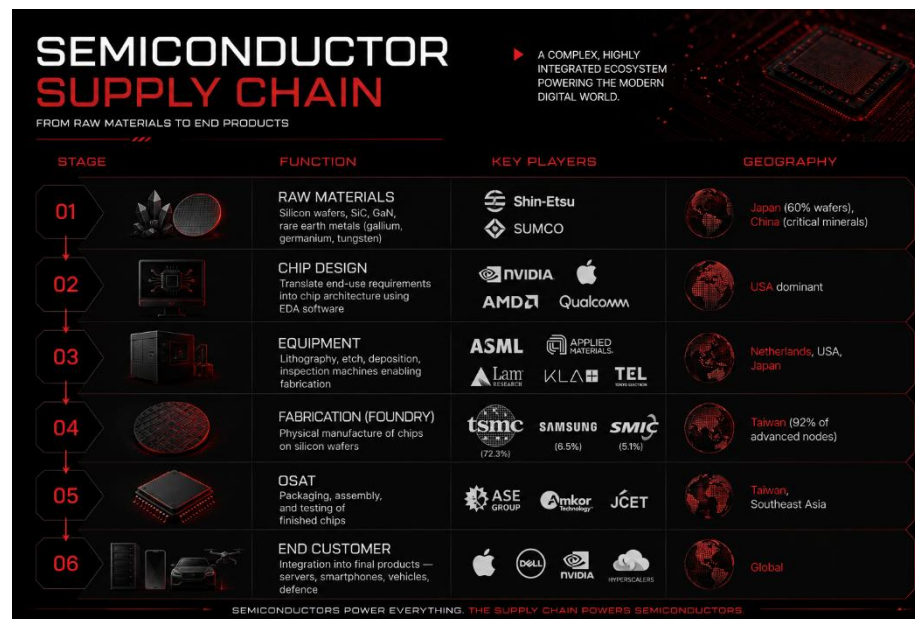
The semiconductor value chain is the process of transforming raw materials into finished chips that can be integrated into various electronic systems. It is highly complex and spread across the globe, involving no single company that controls the entire process rather specialists of various fields like. A detailed breakdown of this chain and its critical chokepoints is covered in the Supply Chain Analysis section.

3 SUPPLY CHAIN ANALYSIS

Semiconductors' supply chain has 6 crucial stages which spread across the globe, with each country specialising in their own segment — raw material suppliers, EDA software providers, fabrication equipment manufacturers, chip designers, foundries, and assembly and test providers. No single company or country controls the entire process, making it one of the most specialised and globally distributed supply chains in any industry.

Key Chokepoints

The visual (Figure 1) the six stages from raw material to end customer, along with the key players and dominant geography at each stage. While the chain functions efficiently under normal conditions, several stages represent critical chokepoints — single points of failure where disruption could ripple across the entire global technology ecosystem.



1) China's Raw Material Dominance (Stage 1 — Raw Materials)

China supplies most critical minerals such as gallium, germanium, and tungsten used throughout chip manufacturing. With no near-term Western alternative supply for these materials, this represents a significant upstream vulnerability sitting at the very start of the value chain.

2) US Control Over EDA Software (Stage 2 — Chip Design)

Synopsys and Cadence, both US companies, control the chip design software that every fabless company and IDM depends on to design a chip in the first place. This gives the United States a powerful and often underappreciated lever of control — without EDA software, advanced chip design simply cannot happen, regardless of where manufacturing occurs.

3) ASML's EUV Monopoly (Stage 3 — Equipment)

ASML is the sole global supplier of EUV lithography machines, which are required for manufacturing any chip at 7nm or below. No alternative supplier exists anywhere in the world, meaning every advanced foundry — including TSMC, Samsung, and Intel — depends entirely on one Dutch company for its most critical piece of equipment.

4) Taiwan's Fabrication Concentration (Stage 4 — Foundry)

Taiwan produces 92% of the world's most advanced semiconductor nodes, with TSMC alone holding 72.3% of global foundry market share. This level of concentration means a single geopolitical event or natural disaster in one location could disrupt the global technology supply chain almost entirely, with no near-term alternative capacity available elsewhere.

Together, these four chokepoints explain why the global semiconductor supply chain, despite being one of the most efficient and specialised systems in industrial history it is also one of the most fragile, with disruption risk concentrated in a small number of companies and countries rather than distributed broadly across the global industry.

4 MARKET SIZE AND GROWTH

The global semiconductor market is expanding at a rapid pace. In 2025, the global market reached \$793 billion USD, and it is estimated to reach approximately \$975 billion by the end of 2026, representing 26% year-on-year growth. The market can be segmented geographically into four regions: Asia-Pacific, the Americas, Europe, and the rest of the world.

Year	Revenue	Growth
2023	\$526.8B	baseline
2024	\$627.6B	+19.1% YoY
2025	\$793B	+21% YoY
2026 (forecast)	~\$975B	+26% YoY
2030 (forecast)	~\$1.3–1.5T	sustained growth
2036 (outlook)	~\$2T	long-term structural

Asia-Pacific: The semiconductor market is geographically concentrated in the Asia-Pacific region, which accounts for the majority of both production and consumption. Major countries such as Taiwan, South Korea, China, and Japan serve as the primary manufacturing hubs. The Asia-Pacific region holds approximately 60% of the total production share of the global market.

Americas: The Americas region specialises in the non-physical side of production — design and IP.

The USA is dominant in the fabless and EDA software segments. The region accounts for 25% of total consumption share of the semiconductor market.

Europe: The European region has specialised in automotive and industrial chips and also dominates in the manufacture of equipment required for chip production. This region holds 10% of the consumption share of the semiconductor market.

Rest of the World: This region includes countries in the Middle East and emerging markets such as

Region	Role	Share of Market
Asia-Pacific	Manufacturing hub — Taiwan, South Korea, China, Japan	~60% of production
Americas	Design and IP hub — USA dominant in fabless and EDA	~25% of consumption
Europe	Automotive and industrial chips — Germany, Netherlands	~10%
Rest of World	Emerging — India (OSAT), Middle East (fab investments)	~5%

India, which is developing as an OSAT hub. This region holds a total of 5% of the global semiconductor market share.

While Asia-Pacific dominates production, the United States captures the largest share of semiconductor industry profits through its dominance in high-value chip design, EDA software, and equipment — despite manufacturing less than 15% of chips physically.

Key Growth Drivers

AI and Generative AI: Generative AI and LLM are the single most powerful demand catalyst the semiconductor industry has ever experienced. AI chips already account for approximately 50% of total industry revenue in 2026, despite representing less than 0.2% of total unit volume — reflecting the extraordinary value concentration. Every new generation of AI model requires exponentially more compute, driving sustained demand for advanced node chips, HBM memory, and high-bandwidth networking silicon. The transition toward Artificial General Intelligence (AGI) represents the next horizon of this demand cycle.

Autonomous Vehicles and EVs: Automotive is the second fastest growing semiconductor end market, at 10.7% CAGR. A conventional vehicle today contains 200–300 semiconductors, while a fully

autonomous vehicle will require more than 1,000. The shift to electric powertrains additionally requires power semiconductors — particularly Silicon Carbide (SiC) and Gallium Nitride (GaN) — for battery management and charging systems.

IoT, Edge AI, and Connected Devices: The proliferation of connected devices across industrial, consumer, and infrastructure applications is creating sustained high-volume demand at mature semiconductor nodes (28nm and above). Edge AI — where artificial intelligence inference runs on the device rather than in a data centre — is accelerating this trend.

Additional Drivers: Defence and, energy transition, and consumer devices collectively support the broader market, even as AI dominates the premium revenue tier.

Demand Trends

Three structural demand trends are reshaping the market beyond simple volume growth.

Value over volume: the industry is shifting from competing on units shipped to competing on value per chip. AI accelerators generating \$30,000–\$40,000 per unit are redefining what semiconductor revenue looks like.

Vertical integration by hyperscale's: Google, Amazon, Microsoft, and Meta are all designing custom silicon for internal AI workloads, manufactured by TSMC. This is creating a new category of demand that bypasses traditional fabless designers.

Advanced packaging as a performance multiplier: as transistor shrinkage approaches physical limits, chiplet architecture and 3D packaging are becoming primary performance drivers, significantly expanding the addressable market for advanced packaging.

5 DEMAND DRIVERS

Semiconductors exist only because applications pull them into demand, so understanding what is driving that pull is a forward-looking map of where the market is heading. The market is forecast to nearly double from \$793B in 2025 to ~\$2 trillion by 2036, with growth concentrating at the high-value leading edge rather than spreading evenly. Four structural drivers explain most of this shift.

AI / Generative AI / LLMs: Training and inference at scale already make AI chips ~50% of 2026 industry revenue from under 0.2% of unit volume the highest value concentration in the industry's history. Datacentres spend is projected to grow at 18% CAGR, from \$156B in 2025 to \$361B by 2030, and the move from current LLMs toward AGI represents a demand multiplier the market hasn't fully priced in yet.

Autonomous Vehicles: Each step up in autonomy multiplies chip content from 200-300 semiconductors in a conventional car to 1,000+ in a Level 5 vehicle. Automotive is already the second-fastest-growing end market at 10.7% CAGR, and with ~90 million vehicles produced annually, even partial adoption adds hundreds of billions in incremental, largely cycle-insulated demand.

Defence & Aerospace: Rising geopolitical tension is driving the largest sustained defence spending increase since the Cold War, and unlike consumer markets, this demand is non-cyclical and price insensitive. The bigger opportunity is technology sovereignty the US, EU, India, Japan, and South Korea are all building semiconductor self-sufficiency into defence budgets, creating a government-backed demand floor independent of private-sector cycles.

Consumer Devices: Still the largest end market by volume (over a trillion chips annually), consumer devices grow slower at 5-6% CAGR but keep mature-node fabs running at the utilisation the rest of the supply chain depends on. The forward opportunity is the AI upgrade cycle - on-device inference in smartphones, AI PCs, and smart home devices - which, across ~6 billion installed smartphones, represents a meaningful incremental demand pulse at both leading and mature nodes.

6 INDUSTRY STRUCTURE & COMPETITIVE LANDSCAPE

Market Concentration

The global semiconductor industry is best described as a tiered oligopoly — highly concentrated at the leading edge of the value chain, and more fragmented at mature nodes and legacy segments. At the foundry level, the market borders on monopolistic, with TSMC alone commanding 72.3% of global foundry revenue as of Q1 2026, making it eleven times larger than its nearest competitor by revenue. At the fabless design level, NVIDIA holds approximately 80–85% of the AI accelerator market, while Synopsys and Cadence together control the EDA software market in a near-duopoly. This concentration is not accidental — it is the structural outcome of an industry where the cost of achieving leadership is so prohibitively high that only a handful of companies can compete at the frontier, while the rest occupy lower-margin, less strategically critical positions further down the value chain.

Competitor Identification Across Four Tiers

The semiconductor competitive universe is structured across four distinct tiers, each defined by business model rather than size alone.

Tier 1 — Pure-play Foundries (Direct Competitors at Manufacturing Layer)

TSMC, Samsung Foundry, SMIC, UMC, GlobalFoundries, and HuaHong all compete for wafer manufacturing contracts from fabless designers. However, only TSMC and Samsung are capable of manufacturing at leading-edge nodes — the rest operate at mature nodes, serving different customer segments entirely.

Tier 2 — IDMs (Indirect Competitors)

Intel and Samsung Electronics design and manufacture their own chips internally, while also offering foundry services externally. Intel Foundry specifically is attempting to re-enter the external foundry market — though it does not appear in the current top ten foundry rankings as of Q1 2026.

Tier 3 — Fabless Designers (Customers, with indirect competitive pressure)

NVIDIA, Apple, AMD, Qualcomm, Broadcom, and MediaTek are TSMC's most important customers rather than competitors. However, their growing move toward custom silicon design — particularly hyperscalers designing their own AI chips — creates a form of indirect competitive pressure on traditional chip designers.

Tier 4 — Ecosystem Enablers (Adjacent, Not Direct Competitors)

ASML, Applied Materials, Lam Research, KLA, Synopsys, and Cadence are critical infrastructure providers. They are not chip manufacturers, but their monopoly or duopoly positions make them among the most strategically powerful entities in the entire industry.

Key Competitive Dynamics

Node leadership is the primary axis of competition at the leading edge. The company that achieves volume production at the smallest node first wins the most valuable customers — NVIDIA, Apple, and AMD all follow the leading node provider. TSMC is currently the only company manufacturing at 2nm and 3nm at commercial scale and yield.

Yield rates separate paper announcements from real competitive capability. Samsung has 3nm process technology but has struggled with yield — the percentage of working chips per wafer — making its 3nm node commercially uncompetitive despite being technically available. TSMC's consistent yield leadership is a primary reason its market share has expanded from 59% in 2023 to 72.3% in Q1 2026 while Samsung's has declined from 13% to 6.5% over the same period.

Customer lock-in is the structural moat that makes competitive dynamics in this industry unusually stable. TSMC's process design kits (PDKs) mean fabless designers optimise their chip architecture specifically for TSMC's manufacturing process — switching foundries would require a full chip redesign, typically taking 18–24 months and hundreds of millions of dollars. This switching cost effectively transforms TSMC's customer relationships into long-term structural partnerships rather than transactional vendor arrangements and is the single most important reason TSMC's market share has remained dominant despite the enormous financial incentives governments and competitors have provided to challenge it.

7 PESTLEA ANALYSIS

PESTLE examines the Political, Economic, Social, Technological, Legal, and Environmental forces shaping an industry. For semiconductors this matters because outcomes are driven less by ordinary competition and more by state policy, geopolitics, and physical constraints on where chips can be made.

Political: Governments now treat semiconductors as a strategic national asset, reshaping global trade and investment flows. This is evident in massive subsidy programmes - the US CHIPS Act (\$52B grants catalysing \$450B private investment) and EU Chips Act (€43B, targeting 20% of global output by 2030) - alongside escalating export controls, with the US restricting chip equipment sales to China and China retaliating with curbs on tungsten, gallium, and germanium. This shows political risk now runs on two fronts at once: a subsidy race to build capacity, and a parallel decoupling effort splitting the industry into a Western-aligned bloc and a self-sufficient Chinese bloc backed by \$150B+ in state funding. Layered on top is the unresolved China-Taiwan tension – Taiwan makes 92% of advanced-node chips, so escalation would trigger the largest supply shock in tech history - while new Section 232 tariffs (25%, Jan 2026) are structurally raising costs industry-wide.

Economical: The industry's extreme capital intensity is now colliding with an unprecedented, AI-driven capex supercycle. A leading-edge 2nm fab costs \$20B+ and takes four to five years to build - only three companies globally can afford it - and TSMC alone is spending \$52-56B on capex in 2026, with total industry capex projected at \$912B between 2024 and 2028. This is driven by AI becoming the structural foundation of demand rather than a temporary catalyst: AI chips already account for ~50% of 2026 revenue from under 0.2% of unit volume, while consumer, automotive, and industrial segments stay weak under tariffs and high rates, creating a two-speed market. The risk is twofold - a boom-bust cycle the AI wave may only be delaying, with overcapacity likely once that capex lands post-2028, and currency exposure unique to Taiwan-based producers (a 1% NTD appreciation cuts TSMC's margins by ~0.4%).

Technological: Technological leadership is now concentrated in a small set of process and packaging breakthroughs, each acting as a chokepoint. TSMC entered 2nm volume production in 2025 and remains the only company at that node at scale, while the next lithography generation depends entirely on ASML's High-NA EUV machines (\$380M+ each), with China permanently excluded from EUV access. Beyond node shrinkage, advanced packaging (TSMC's CoWoS, doubling capacity through 2026 to address a \$40B+ market) and HBM (led by SK Hynix, +37% revenue in 2025) are equally critical - NVIDIA's top GPUs depend on CoWoS and HBM as much as on the node itself. This means competitive advantage now rests on an interlocking set of chokepoints rather than one technology, each also doubling as a lever of export control, as seen with HBM's recent addition to the restricted list (full detail in the Technology Trends section).

Social: The concentration of semiconductor engineering talent in Asia is increasingly mismatched with where new fab capacity is being built. China, Japan, South Korea, and Taiwan produce most of the world's semiconductor talent, and this has not followed capacity to the US, Europe, and India; the US alone faces a projected shortfall of 67,000 engineers and technicians by 2030 (SIA/Deloitte), and TSMC's Arizona project was delayed partly for this reason. Demand-side trends reinforce growth regardless - with ~6 billion smartphones in the global base now entering a partial AI upgrade cycle, even modest adoption adds significant incremental chip demand. But this does not remove the supply-side talent constraint, and rising community opposition to new fabs - as TSMC faced in Arizona over water and land use - means social factors now shape both how fast and where capacity can be built.

Legal: Legal exposure centres on three areas - intellectual property, export control compliance, and the conditions attached to government funding. IP remains the industry's primary moat, with litigation concentrated in advanced packaging and chiplet interconnects; separately, the US Foreign Direct Product Rule (FDPR) gives Washington extraterritorial reach over any product made anywhere using US-origin technology, now covering EDA software, AI model weights, HBM, and packaging equipment. This is compounded by binding CHIPS Act guardrails - recipients including TSMC, Intel, Samsung, and Micron face a 10-year prohibition on expanding advanced capacity in China and profit-sharing obligations above certain return thresholds. Together this mean accepting government support now comes with multi-year strategic constraints, and industry bodies have already warned these guardrails are impeding the cross-licensing activity the industry depends on.

Environmental: Environmental constraints on fabs are best understood as direct business risks - to continuity, cost, and timelines - not purely ecological concerns. Fabs use ~5M gallons of ultra-pure water daily, a serious issue given Arizona and Texas are water-stressed and Taiwan's 2021 drought forced TSMC to truck in water; separately, tightening EU/US F-gas regulation is directly raising compliance costs for the high-warming gases used in fabrication. These pressures compound with hazardous waste rules adding permitting delays (as seen at TSMC Arizona) and with energy exposure - TSMC alone uses 8% of Taiwan's electricity, exposing it to both cost and ESG pressure as it works toward an unmet 2050 renewables target. Geography intensifies this further: Taiwan sees 1,000+ earthquakes a year (the April 2024 quake cost TSMC \$92.4M), and PwC estimates a third of global chip output faces climate disruption risk within a decade - a recurring, quantifiable risk line, not a background concern.

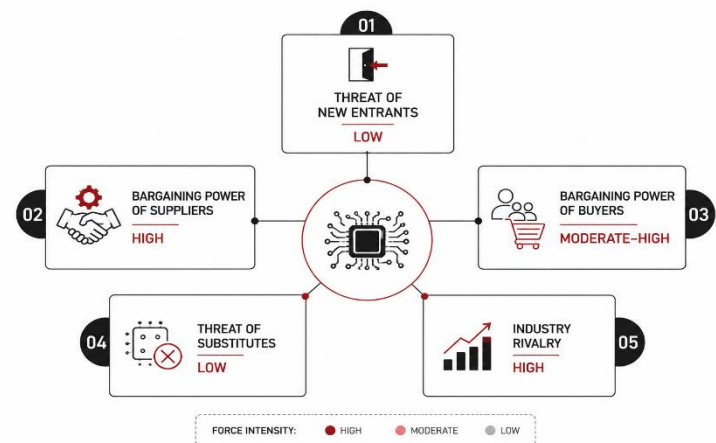
8 PORTER'S FIVE FORCES

1. Threat of New Entrants (Low): The semiconductor industry has very high entry barriers. The industry requires high capital, technological and manufacturing expertise, IP and patents, as well as years of R&D and expert talent. Setting up new fabs requires capital of up to \$20 billion, and it takes extremely long for them to become profitable. Highly advanced EDA software and EUV machines are required for the design and manufacturing process. Existing players have strong patents and licensing policies to keep competition out of the game. There is also a huge shortage of talented workforce — it is estimated that demand for 67,000 additional engineers and technicians will be required in the US alone by 2030. In the end, it can be concluded that there is very low possibility of new players entering the industry.

2. Bargaining Power of Suppliers (High): Supplier power in the semiconductor industry is high due to the concentration of critical suppliers. Companies such as TSMC dominate advanced semiconductor manufacturing, producing most of the world's leading-edge chips, while ASML is the sole supplier of EUV lithography machines required for advanced process nodes. The industry also depends on a limited number of suppliers for specialty chemicals, wafers, semiconductor equipment, and rare-earth materials. High switching costs, long qualification cycles, and supply chain concentration give suppliers considerable bargaining power, particularly during periods of strong demand.

3. Bargaining Power of Buyers (Moderate):

Buyer power is moderate. Only a few companies supply advanced chips, and switching foundries involves high costs, lengthy qualification processes, and redesign risks, while demand continues to exceed supply—limiting buyer leverage. However, major customers such as Apple, NVIDIA, and AMD account for a significant share of suppliers' revenue (TSMC's top ten customers contribute over 76% of net revenue), giving them meaningful negotiating power over pricing and capacity allocation. Overall, high switching costs constrain buyer power, while the scale and strategic importance of large customers strengthen it, resulting in moderate bargaining power.



4. Threat of Substitutes (Low): There are no direct substitutes for semiconductors available in the near future. They are required in almost every electronic product we use, and no alternative material currently matches silicon's stability and reliability at room temperature for large-scale chip manufacturing. In the far future, quantum computing could disrupt parts of the industry, but a meaningful impact is unlikely for at least a few decades. Some software optimisation can reduce hardware demand at the margins, but only marginally. Ultimately, it can be concluded that there are no real substitutes available for semiconductors in the foreseeable future.

5. Industry Rivalry (High): Existing players are highly interdependent but also compete intensely with one another. Companies continuously invest in R&D to innovate and keep rivals out of the game by patenting their IP, with the US leading globally in patent submissions and issuance as clear evidence of this race. New-generation technology is launched frequently to compete on performance, power

efficiency, and manufacturing supremacy. Even with TSMC's overwhelming dominance, the broader foundry market still grew 30% year-on-year in Q1 2026, showing that competitors like Samsung and Intel continue to invest heavily to capture their share of AI-driven demand. To sum it all up, there is high rivalry between industry players to get ahead of one another and bring stronger products to market.

9 RISK ANALYSIS

The risks identified across the PESTLE analysis, Porter's Five Forces, and competitive landscape sections are consolidated below into a single reference matrix. Detailed explanations of each risk are covered in their respective sections — this matrix serves as a structured summary for investment decision-making purposes.

Risk	Category	Likelihood	Mitigation
Taiwan military conflict or blockade	Geopolitical	Medium	TSMC Arizona (2027) and Japan fabs reduce but do not eliminate concentration
China critical materials export restrictions	Geopolitical	High	Western nations building alternative supply chains for gallium, germanium, tungsten
AI demand concentration — ~50% of revenue from one end market	Financial	Medium	Automotive, defence, and consumer diversification providing secondary demand floor
Semiconductor overcapacity post-2028	Financial	Medium	Long-term hyperscaler supply agreements locking in leading-edge demand

Overall risk profile: The semiconductor industry carries a concentrated and geopolitically driven risk profile — the majority of high-severity risks trace back to Taiwan's geographic concentration and the US-China technology conflict rather than to demand fundamentals or competitive dynamics. This asymmetry is important for investors: the industry's commercial outlook is strong, but the tail risks are large and politically determined rather than market-determined, making them difficult to hedge through conventional portfolio management.

10 SWOT ANALYSIS

Strengths: The semiconductor industry has strong long-term competitive advantages because AI chips now generate around 50% of industry revenue, governments are investing billions through semiconductor incentive programs, and extremely high barriers to entry protect established companies. These factors create stable demand, encourage continuous innovation, and limit new competition, allowing leading firms to maintain strong market positions. As a result, the industry is well positioned for sustainable long-term growth and profitability.

Weaknesses: Despite its growth potential, the semiconductor industry faces significant challenges due to its heavy dependence on Taiwan for advanced chip production, recurring boom-bust cycles, and a global shortage of skilled workers. These issues increase supply chain risks, delay capacity expansion, and expose companies to periods of overcapacity and lower profitability. Therefore, managing operational and supply chain risks remains critical for the industry's long-term stability.

Opportunities: Rapid advances in artificial intelligence, autonomous vehicles, defence electronics, and advanced packaging technologies are creating major growth opportunities for the semiconductor industry. These expanding markets require increasingly powerful and specialised chips, driving higher demand and encouraging further investment in innovation and manufacturing capacity. Consequently, the industry has multiple long-term drivers that support continued revenue and market expansion.

Threats: The semiconductor industry faces growing external risks from geopolitical tensions, export restrictions on critical minerals, potential post-2028 oversupply, and the long-term emergence of quantum computing. These factors could disrupt global supply chains, increase costs, reduce profitability, and create uncertainty for future investment decisions. Therefore, companies must effectively manage geopolitical and technological risks to maintain long-term competitiveness.

SWOT ANALYSIS			
STRENGTHS		WEAKNESSES	
	AI chips now account for around 50% of industry revenue, creating a strong long-term demand base that is driven by structural growth, not short-term cycles.		Around 92% of advanced chips are produced in Taiwan, creating the largest geographic concentration risk in any global industry.
	TSMC holds 72.3% of the global foundry market at advanced nodes, giving it a near-monopoly position with no credible challenger in the near term.		The semiconductor industry has a history of 3–4 year boom-bust cycles. Overcapacity risk increases as \$912B of global capex comes online by 2028.
	Governments are investing heavily through the US CHIPS Act (\$52B), EU Chips Act (€43B), and Japan incentives (\$26B), which reduces risk and supports private investment.		There is a global engineering talent shortage, with an estimated 67,000 technician shortfall in the US alone by 2030, which constrains fab expansion timelines.
	The industry faces extremely high barriers to entry. Fabs cost \$20B+ to build, require ASML equipment, and take years of expertise to achieve high yields, protecting incumbents.		Semiconductor fabs consume 8% of Taiwan's electricity and 5 million gallons of water daily. This high resource dependence creates regulatory and operational risks.
OPPORTUNITIES		THREATS	
	AGI development requires exponentially more compute power. Each new model generation multiplies demand for advanced nodes, driving long-term industry growth.		China is weaponising critical mineral exports. Tungsten exports will be restricted from February 2026, while gallium and germanium have been restricted since 2023, threatening supply chains.
	Autonomous vehicles are expected to scale from 200–300 chips per vehicle today to 1,000+ chips at full autonomy, supporting a 10.7% CAGR in semiconductor demand from this end market.		US-China geopolitical escalation, including export controls, tariffs, and Taiwan military tension, could create a structural bifurcation in the semiconductor supply chain.
	Defence and aerospace spending provides a stable, non-cyclical and price-insensitive demand floor driven by government procurement and national sovereignty.		Overcapacity after 2028 is a major risk. With \$912B of global capex coming online simultaneously, the market could face a supply glut if AI demand moderates.
	Advanced packaging technologies such as CoWoS, chiplets, and HBM stacking are becoming as strategically important as node shrinkage. This \$40B+ market is growing rapidly and creating new revenue opportunities.		Quantum computing is a long-term architectural disruptor. While decades away from commercial scale, it could eventually reduce demand for traditional semiconductors.

11 TECHNOLOGY TRENDS

Five technological advancements are currently reshaping the competitive landscape of the semiconductor industry, each functioning less as a scientific milestone and more as a direct driver of revenue concentration, customer lock-in, and market share.

TSMC's **2nm node (N2)**, entering volume production in 2025, is the most commercially significant advancement — it is the only leading-edge node available at commercial scale and yield, meaning every major fabless designer requiring maximum performance must use TSMC, directly reinforcing its 72.3% foundry market share. **High-NA EUV**, ASML's next-generation lithography machine now entering fabs, determines who leads the node generation after 2nm — whoever masters it first controls the roadmap for the rest of the decade, and TSMC's early access gives it a structural head start over Samsung and Intel. **CoWoS advanced packaging** has become as strategically important as node shrinkage — TSMC is doubling CoWoS capacity in 2025–2026 to address a \$40B+ market, driven entirely by NVIDIA's Blackwell GPU requirements, which physically cannot exist without this packaging technology. **HBM memory**, led by SK Hynix with 37% revenue growth in 2025, is the performance enabler that makes AI chips viable at scale — stacked directly onto AI accelerators to deliver the bandwidth that large language models require, and now subject to US export controls to China, confirming its status as a strategic rather than merely commercial technology. Finally, **GAA transistors**, replacing FinFET architecture at sub-3nm nodes, represent the fundamental shift in how transistors are physically constructed — TSMC's implementation at 2nm delivering better power efficiency and performance, and further widening the process technology gap between TSMC and its foundry competitors.

All five trends converge on a single investment conclusion: the companies positioned at the intersection of these advancements — primarily TSMC as manufacturer, NVIDIA as the leading chip designer, and SK Hynix in memory — are capturing a disproportionate and growing share of the industry's total economic value. This sets the analytical foundation for the TSMC-specific deep dive in Part 2.

12 INDUSTRY FINANCIAL METRICS

The semiconductor industry's financial profile varies dramatically by business model, making industry-wide averages misleading without segment-level context. A fabless company and a foundry operating in the same industry can show gross margins that differ by 20 percentage points — not because one is performing better, but because their cost structures are fundamentally different by design.

Metric	Fabless (NVIDIA)	Foundry (TSMC)	IDM (Intel)
Gross Margin	75.0%	59.9%	34.8%
Operating Margin	62.4%	50.8%	-4.2%
EBITDA Margin	~63–65%	~65–68%	~15–18%
Net Margin	~55%	45.0%	Negative
Capex % of Revenue	<3%	~35%+	~20–25%
D/E Ratio	0.11	0.19	Higher leverage

The logic behind these differences is straightforward. Fabless companies like NVIDIA carry no manufacturing assets on their balance sheet — their capital goes into R&D and sales rather than fabs — which produces the highest margins and lowest capex intensity in the industry. Foundries like TSMC absorb enormous capital expenditure (\$52–56B in 2026 alone) but earn it back through scale across hundreds of customers, producing strong but lower margins. IDMs like Intel carry both the cost burden of manufacturing and the competitive pressure of design — currently producing the weakest financial results of any segment as Intel loses market share on both fronts simultaneously.

Why ROCE matters most here: In a capital-intensive industry where a single fab costs \$20B+, return on capital employed is the most revealing profitability metric — it shows whether companies are generating adequate returns on the enormous physical assets they deploy. A high gross margin means little if the capital required to generate it is destroying shareholder value. For foundries and IDMs specifically, ROCE is the metric that best captures whether scale and utilisation are translating into genuine economic returns

13 FUTURE OUTLOOKS

The semiconductor industry's trajectory over the next decade is one of the clearest structural growth stories in any global market — but it is not without meaningful inflection points and genuine uncertainties that will separate the winners from the casualties.

Short-Term (2026–2027)

The immediate picture is one of a market running at full speed on one engine while the other sputters. The global market is on course to reach approximately \$975 billion in 2026, driven almost entirely by AI data centre buildout rather than broad-based demand recovery. Memory prices are projected to spike sharply by mid-2026, which will suppress consumer electronics demand even as hyperscaler AI spending remains resilient. The most significant structural development of this period is TSMC Arizona Fab 2 targeting 3nm production in late 2027 — the first meaningful advanced node capacity to exist outside Taiwan, marking a genuine milestone in supply chain diversification even if it represents a small fraction of total global capacity. Export controls will continue tightening, with EDA software, HBM memory, and AI model weights all newly restricted — accelerating the bifurcation of the industry into two increasingly separate technology ecosystems.

- Automotive semiconductor content scaling significantly as autonomous driving levels advance — 10.7% CAGR end market becoming a major revenue contributor
- India establishing itself as a credible OSAT hub — Tata Dholera (28nm) and Micron Gujarat both operational

Long-Term (2030–2036)

Looking further out, the industry's path to \$2 trillion in annual revenue by 2036 is credible but not guaranteed — it requires sustained 7.5% CAGR from current levels, which depends on several demand categories beyond AI maturing simultaneously. Advanced packaging becomes as economically significant as fabrication itself. Humanoid robotics emerges as a meaningful new demand category, with motion control, AI inference, and sensor fusion chips scaling alongside robot production volumes. Quantum computing begins narrow commercial deployment — not displacing classical semiconductors but creating an adjacent demand category. Most significantly, if AGI is achieved within this window, the computational requirements it would generate are beyond any current projection and would represent the single largest demand multiplier the industry has ever experienced.

Key Uncertainties

Three questions will determine whether this outlook materialises or requires significant revision. First, whether AI demand is structural or cyclical — the answer to this single question determines whether 2028 brings continued growth or a severe overcapacity correction that reshapes the competitive landscape entirely. Second, Taiwan's geopolitical stability — the one variable that no amount of capex, subsidy, or supply chain planning can fully hedge within this decade. Third, the speed of China's domestic semiconductor development — if state-backed players close the technology gap faster than current export controls anticipate, the competitive and geopolitical dynamics of the entire industry shift in ways that are difficult to model from today's vantage point.

14 CONCLUSION

The global semiconductor industry is one of the strongest long-term growth industries, with the market expected to grow from **\$793 billion in 2025** to nearly **\$2 trillion by 2036**. This growth is mainly driven by the increasing use of AI, cloud computing, data centres, autonomous vehicles, and other digital technologies. The industry also has very high barriers to entry, requiring billions of dollars of investment, advanced manufacturing technology, and years of technical expertise. As a result, leading companies such as TSMC and NVIDIA continue to hold strong market positions, while government support through semiconductor incentive programs is helping expand global manufacturing capacity.

Despite these strengths, the industry's biggest risks come from geopolitics rather than normal market conditions. Around **92% of advanced chips are manufactured in Taiwan**, making the global supply chain highly dependent on a single region. Rising tensions between the US and China, export restrictions on critical minerals, and the possibility of oversupply as new manufacturing capacity comes online could affect future growth and profitability. Overall, the semiconductor industry remains highly attractive for long-term investment because of its strong demand and competitive advantages, but investors should also consider the geopolitical and supply chain risks that could influence future performance.

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